

## Yiran Wang

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| CONTACT<br>INFORMATION  | 601-85 Bloor St E<br>Toronto, ON<br>Canada M4W 3Y1                                                                                                                                                           | <i>Phone:</i> (548) 989-4168<br><i>Email:</i> yiranwang.wang@utoronto.ca<br><i>Personal Website</i> |
| RESEARCH<br>INTERESTS   | Bayesian Inference, Capture-Recapture, Causal Mediation Analysis, Latent Factor Modeling,<br>Data Integration, Statistical Ecology                                                                           |                                                                                                     |
| ACADEMIC<br>AFFILIATION | <b>Postdoctoral Fellow</b>                                                                                                                                                                                   | May 2026 - Apr 2028                                                                                 |
|                         | Institute of Health Policy, Management and Evaluation, University of Toronto<br>Advisor: Kuan Liu, Ph.D.                                                                                                     |                                                                                                     |
|                         | <b>Postdoctoral Associate</b>                                                                                                                                                                                | Nov 2024 - Oct 2025                                                                                 |
|                         | Dept. of Biostatistics, Yale School of Public Health<br>Topic: <i>Causal Mediation Analysis on Exposure Mixtures with Latent Factor Models</i><br>Advisor: Bhramar Mukherjee, Ph.D.                          |                                                                                                     |
|                         | <b>Ph.D. in Statistics</b>                                                                                                                                                                                   | Sep 2019 - Aug 2024                                                                                 |
|                         | Dept. of Statistics and Actuarial Science, University of Waterloo<br>Thesis: <i>Modeling and Bayesian Computations for Capture-Recapture Studies</i><br>Advisors: Audrey Béliveau, Ph.D., Martin Lysy, Ph.D. |                                                                                                     |
|                         | <b>M.S. in Statistics</b>                                                                                                                                                                                    | Sep 2017 - Jun 2019                                                                                 |
|                         | Dept. of Statistics, University of California Irvine                                                                                                                                                         |                                                                                                     |
|                         | <b>B.S. in Statistics</b>                                                                                                                                                                                    | Sep 2013 - Jun 2017                                                                                 |
|                         | Dept. of Mathematics, Donghua University<br>Thesis: <i>Influence of China's Innovation on Total Factor Productivity Based on Quantile Regression</i><br>Advisor: Rong Jiang, Ph.D.                           |                                                                                                     |
| HONORS &<br>AWARDS      | Canada Impact+ Research Training Awards (\$70,000/year)                                                                                                                                                      | May 2026 - Apr 2028                                                                                 |
|                         | 2026 ISBA World Meeting Junior Travel Award                                                                                                                                                                  | Apr 2026                                                                                            |
|                         | 2025 Joint Statistical Meetings Early Career Travel Award                                                                                                                                                    | Jun 2025                                                                                            |
|                         | Statistical Society of Canada Annual Meeting Student Grant                                                                                                                                                   | May 2024                                                                                            |
|                         | University of Waterloo Doctoral Thesis Completion Award                                                                                                                                                      | Apr 2024                                                                                            |
|                         | Women in Mathematics Directed Reading Program Distinguished Service Award                                                                                                                                    | Apr 2024                                                                                            |
|                         | Women in Mathematics Directed Reading Program Mentorship Award                                                                                                                                               | Aug 2023, Apr 2024                                                                                  |
|                         | University of Waterloo Graduate Student Research Dissemination Award                                                                                                                                         | May 2023, Mar 2024                                                                                  |
|                         | The Eleventh Annual Canadian Statistics Student Conference Travel Award                                                                                                                                      | May 2023                                                                                            |
|                         | University of Waterloo Graduate Studies Conference Assistantship                                                                                                                                             | May 2022                                                                                            |
|                         | University of Waterloo Senate Graduate Scholarship                                                                                                                                                           | Mar 2022                                                                                            |
|                         | Waterloo Special Graduate Student Entrance Award                                                                                                                                                             | Sep 2019                                                                                            |

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|----------------------------------------------------------------------------|--------------------|
| Donghua University Outstanding Graduate                                    | Jun 2017           |
| Donghua University Merit Scholarship                                       | Dec 2016           |
| Donghua University Tianji Scholarship                                      | Dec 2016           |
| Donghua University Social Work Excellence Scholarship                      | Dec 2015, Dec 2016 |
| Donghua University Outstanding Student Leader                              | Dec 2015, Dec 2016 |
| 1 <sup>st</sup> Prize in Donghua University Mathematical Modelling Contest | Jun 2015           |

- PUBLICATIONS
1. Maxwell Salvatore, **Yiran Wang**, Preeti Syal, Brian Wahl, Harsh Parikh, Vishal Deo, Naveen Kumar Bhatraju, Kaushalya Jayaweera, Nisha Rana, Redoy Ranjan, Fyezah Jehan, Abdullah Yusuf, Karthik Adapa, Mona Duggal, Anurag Agrawal, and Bhramar Mukherjee (2026). The Case for an Integrated Biobanking Initiative in South Asia. *The Lancet Regional Health - Southeast Asia*. 49: 100777 [DOI]
  2. Sean McGrath<sup>†</sup>, **Yiran Wang**<sup>†</sup>, Yi-Ting Lin, John D. Meeker, Sung Kyun Park, Joshua L. Warren, and Bhramar Mukherjee (2026). A Comparison and Evaluation of Statistical Methods for Mediation Analysis with Mixtures of Environmental Exposures. *BMC Medical Research Methodology*. 26: 93 [DOI]
  3. **Yiran Wang**, Alicia E. Boyd, Lillian Rountree, Yi Ren, Kate Nyhan, Ruchit Nagar, Jackson Higginbottom, Megan L. Ranney, Harsh Parikh, and Bhramar Mukherjee (2026). Ten Core Concepts for Ensuring Data Equity in Public Health. *JAMA Health Forum*. 7(1): e256031. [DOI]
  4. Daniel Y. Dapaah, Shoutaro Arakawa, Gwennyth A. Carroll, **Yiran Wang**, Stewart McLachlin, Mitsuru Saito, and Thomas L. Willett (2026). Advanced Glycation End-product Adducts Alter the Bone Collagen Network and Human Cortical Bone Fracture Resistance. *JBMR Plus*. 10(1): ziaf172. [DOI]
  5. **Yiran Wang**, Martin Lysy, and Audrey Béliveau (2025). Plant-Capture Methods for Estimating Homeless Population Size From Uncertain Plant Captures. *Epidemiology*. 36(3): 319–326.
  6. Lauren Raymond-King, Sean McGrath, Bhramar Mukherjee, **Yiran Wang**, Harsh Parikh, John Rothen, Pamela R. Soulos, Cary P. Gross, and John W. Kunstman. Frailty and Low Utilization of Curative-Intent Surgical Resection in Non-Metastatic Pancreatic Cancer. *JNCI Cancer Spectrum*. Revised and resubmitted.[DOI]
  7. **Yiran Wang**, Martin Lysy, and Audrey Béliveau. Rapid Scaling of Compositional Uncertainty from Sample to Population Levels. *Statistical Methods & Applications*. Under review. [arXiv]
  8. Daniel Y. Dapaah, Shoutaro Arakawa, Gwennyth A. Carroll, **Yiran Wang**, Stewart McLachlin, Mitsuru Saito, and Thomas L. Willett. Is direct oxidative damage of bone collagen associated with poor human cortical bone tissue fracture resistance? *JBMR Plus*. Under review.
  9. **Yiran Wang**, Alicia E. Boyd, Lillian Rountree, Yi Ren, Kate Nyhan, Ruchit Nagar, Jackson Higginbottom, Megan L. Ranney, Harsh Parikh, and Bhramar Mukherjee (2025). Towards Enhancing Data Equity in Public Health Data Science. *arXiv preprint arXiv:2508.20301* [arXiv]
  10. **Yiran Wang**, Yi-Ting Lin, Sean McGrath, John D. Meeker, Sung Kyun Park, Joshua L. Warren, and Bhramar Mukherjee. A Tutorial on Conducting Mediation Analysis with Exposure Mixtures. *arXiv preprint arXiv:2509.10916* [arXiv]
  11. **Yiran Wang**, Martin Lysy, and Audrey Béliveau (2024). Method of Moments Estimation of the Bayesian Credible Interval for Markov Chain Monte Carlo. *Working Paper*.

<sup>†</sup> Co-first authorship

# Curriculum Vitae

## ADDITIONAL PROFESSIONAL EXPERIENCE

### Research Assistant

Jan 2020 - Apr 2024

*University of Waterloo, Waterloo, ON*

- Built a new framework for the plant-capture method to estimate population size by incorporating uncertainty in the plants' assessments and also other factors with Bayesian and frequentist inference approaches.
- Analyzed the stock population, proposed new Bayesian genetic mark-recapture methods, and applied to simulation studies and real-data analysis.
- Advisors: Audrey Béliveau, Ph.D., Martin Lysy, Ph.D.

### Consultant

Sep 2023 - Apr 2024

*UW Statistical Consulting & Survey Research Unit, Waterloo, ON*

- Advising students and faculty members on how to leverage sound statistical approaches in their analyses, with more active involvement in enhanced service projects.
- Advisor: Joslin Goh, Ph.D.

## TEACHING EXPERIENCE

### Teaching Assistant

*University of Waterloo, Waterloo, ON*

Sep 2019 - Apr 2024

- Held regular office hours, proctored tests, designed and marked assignments and tests for:
  - STAT 202 Introductory Statistics for Scientists Fall 2019
  - STAT 230 Probability Winter 2020, Spring 2020
  - STAT 330 Mathematical Statistics Winter 2021, Fall 2021
  - STAT 331 Applied Linear Models Spring 2020, Fall 2020, Winter 2021
  - STAT 371 Applied Linear Models and Process Improvement for Business Fall 2019, Fall 2020, Winter 2022, Fall 2022
  - STAT 431 Generalized Linear Models and their Applications Spring 2022, Spring 2023
  - STAT 436/836 Introduction to the Analysis of Spatial Data in Health Research Winter 2023, Winter 2024

*University of California, Irvine, Irvine, CA*

Apr 2018 - Mar 2019

- Held regular office hours, conducted weekly discussion sections, graded homework and examinations for:
  - STATS 7 Basic Statistics Winter 2019
  - STATS 67 Introduction to Probability and Statistics for Computer Science Spring 2018, Fall 2018

## CONFERENCE TALKS

### Bayesian Mediation Analysis with Latent Exposure Mixtures

2025 Joint Statistical Meetings

Aug 2025

### Plant-Capture Methods Using Uncertain Plant Captures

The 2026 New Researchers Conference

Jul 2026

The 2026 ISBA World Meeting

Jun 2026

The 38th New England Statistics Symposium

Jun 2025

The Bayesian Young Statisticians Meeting 2025

Apr 2025

The 2025 ENAR Spring Meeting

Mar 2025

The Statistical Society of Canada 2024 Annual Meeting

Jun 2024

# Curriculum Vitae

2023 Waterloo Student Conference in Statistics, Actuarial Science and Finance  
Oct 2023

## Genetic Mark-Recapture Methods for Stock Populations

2023 Joint Statistical Meetings Aug 2023  
The Statistical Society of Canada 2023 Annual Meeting May 2023  
EURING Analytical Meeting & Workshop 2023 Apr 2023  
The 2022 ISBA World Meeting Jun 2022  
International Statistical Ecology Conference 2022 Jun 2022  
The Statistical Society of Canada 2022 Annual Meeting Jun 2022  
The Statistical Society of Canada 2021 Annual Meeting Jun 2021  
EURING Analytical Meeting & Workshop 2021 Jun 2021

## SERVICE

### Journal Reviewer

American Journal of Epidemiology, Statistical Methods & Applications

### Event and Organization

Instructor & Program Coordinator Mar 2025 - Jul 2025  
*2025 Big Data Summer Immersion at Yale (BDSY)*

Committee Member May 2023 - Apr 2024  
*University of Waterloo Graduate International Student Advisory Committee (ISAC-G)*

Conference Organizing Committee Sept 2022 - May 2023  
*The Eleventh Annual Canadian Statistics Student Conference (CSSC)*

Conference Organizing Volunteer Nov 2021  
*The 2nd Waterloo Student Conference in Statistics, Actuarial Science and Finance*

Student Representative Oct 2018 - Apr 2019  
*Department of Statistics, University of California Irvine*

### Mentorship

Mentor Jan 2024 - Apr 2024, May 2023 - Aug 2023  
*University of Waterloo Women in Mathematics Directed Reading Program*

Mentor Volunteer Apr 2021, Apr 2022, Apr 2023, Apr 2024  
*ASA DataFest*

## PROFESSIONAL DEVELOPMENT

Intermediate Teaching Workshops: *Participation and Equity* 2025  
Yale University

Intermediate Teaching Workshops: *Classroom Observation* 2025  
Yale University

Intermediate Teaching Workshops: *Teaching International Students* 2025  
Yale University

Advanced Teaching Workshops: *Lecturing for Active Engagement* 2025  
Yale University

Intermediate Teaching Workshops: *Leading Effective Discussions* 2025  
Yale University

## Curriculum Vitae

Canadian Statistical Sciences Institute CRT Workshop: *Modern Methods in Survey Sampling*  
University of Ottawa 2024

Canadian Ecological Forecasting Initiative Summer School: *Forecasting for Decision-Making: An Epidemiological & Ecological Perspective* 2023  
The Fields Institute for Research in Mathematical Sciences

Workshop: *NIMBLE virtual short course* 2023  
NIMBLE Development Team

Canadian Statistical Sciences Institute CRT Summer School: *Modern Methods in Survey Sampling*  
University of Ottawa 2022